Tae Hyeon Kweon

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Education

University of Waterloo Ontario, Canada
MASc in Mechanical and Mechatronics Engineering | GPA: 88.5 Sep. 2024 – Present
Advisor: Prof. Soo Jeon
Relevant Coursework: Optimal Control, Adaptive Control, Video Processing, Machine Learning

The Hong Kong Polytechnic University (HKPolyU) Kowloon, Hong Kong
B.Eng. (Honours) in Mechanical Engineering Sep. 2018 – Jun. 2024
Graduated with First Class Honors (Mandatory military service: 2021–2023)
Thesis: *A Mobile Robot with an Active Suspension System for Navigation in Uneven Terrain*

Publications

Tae Hyeon Kweon, Soo Jeon. *Semantic-Aware Active Perception for Next-Best-View Grasp Planning.* International Journal of Precision Engineering and Manufacturing (IJPEM), 2026, In Press, `https://hdl.handle.net/10012/22970`

Tae Hyeon Kweon, Soo Jeon. *ViewScorer: Distilling Vision-Language Models for Efficient Next-Best-View Selection in Robotic Grasping.* Manuscript in preparation, 2026.

Research & Work Experience

Mechanical Systems Control Lab, UWaterloo Ontario, Canada
Graduate Student Researcher Sep. 2024 – Present
Advisor: Prof. Soo Jeon
– Developed a semantic Next-Best-View framework for grasp planning under occlusion, integrating semantic segmentation, volumetric mapping, and grasp synthesis.
– Developed a VLM-distilled policy that predicts the Next-Best-View for grasping via a lightweight network, achieving 2.6× faster inference with comparable success to information gain baselines.

Robotics and Machine Intelligence Lab, HKPolyU Kowloon, Hong Kong
Undergraduate Research Assistant Dec. 2020 – Jun. 2021
Advisor: Prof. David Navarro-Alarcon
– Built a mobile robot capable of autonomous navigation over liquid and muddy terrain.
– Implemented a vision-based localization and navigation system using artificial markers.
– Contributed to securing project funding.

Origami Labs (Start-up) Tsuen Wan, Hong Kong
Engineering Intern Jul. 2020 – Sep. 2020
– Integrated an accelerometer-based trigger system for automatic microphone activation.
– Applied signal-processing filters to reduce environmental noise in wearable prototypes.

Selected Projects

VLA Policy Evaluation System Feb. 2026 – Apr. 2026
Personal Project
– Collected robot manipulation trajectory data via Meta controller-based teleoperation and built a training pipeline.
– Fine-tuned a $\pi_{0.5}$-based VLA model on the collected demonstrations.
– Implemented an automated grasp success evaluation pipeline using rule-based prompting and Qwen2.5-VL.

End-to-End Visual Grasping with PPO, UWaterloo Jan. 2025 – Apr. 2025
SYDE 673 Video Processing and Analysis (Course Project)
– Developed and trained a PPO-based visuomotor grasping policy using the Barrett WAM arm in the Genesis simulator.
– Designed a visual reward function based on corner overlap, contact detection, and lift height.
– Built a multi-environment camera system for robust, parallelized vision-based training.

Adaptive Control of a 2-DOF Robotic Arm, UWaterloo Sep. 2024 – Dec. 2024
ME780 Adaptive Control (Course Project)
– Implemented a Model Reference Adaptive Control (MRAC) scheme to adaptively estimate unknown dynamics and ensure stable trajectory tracking.
– Derived Lyapunov-based adaptation laws and validated controller performance in MATLAB/Simulink.

Active Suspension for Navigating Rough Terrain, HKPolyU Sep. 2023 – Jun. 2024
Undergraduate Thesis
– Designed a mobile robot with four independently actuated suspension units driven by stepper motors.
– Integrated IMU feedback to detect terrain pitch and actively adjust body height for stable navigation improving robustness on uneven terrain.

Teaching Experience

ME549/MTE544 Autonomous Mobile Robotics, University of Waterloo
Teaching Assistant Winter 2025, Fall 2025, Winter 2026
– Led lab sessions on localization, control, and planning; held office hours; graded assignments and exams.

Scholarships & Awards

International Master's Award of Excellence (IMAE), UWaterloo 2024 – 2026
Dean's Honour List, HKPolyU 2021
Full Entry Scholarship, HKPolyU 2018 – 2024

Extracurricular Activities

Volunteer (Score Tally), MME Graduate Research Symposium, UWaterloo Nov. 2024
Peer Tutor in AMA1120 (Basic Mathematics II), HKPolyU Jan. 2024 – May 2024
Mentor, Korean Student Association, HKPolyU Sep. 2023 – Dec. 2023
Team Leader, STEM Learning Kits for Overseas Students, HKPolyU Jun. 2020 – Aug. 2020

Skills

Programming:	Python, C++, MATLAB, Shell
Tools:	PyTorch, ROS 1 & 2, OpenCV, Open3D, Git, Docker, LaTeX
Robots:	Franka Panda, Barrett WAM, TIAGO++, TurtleBot
Simulation:	PyBullet, Genesis, Gazebo, Simulink
Hardware:	IMU, RealSense L515/D455, LiDAR, Arduino, Jetson Orin, Raspberry Pi